

Quota and Strong Population Monotonicity Are Incompatible at Fixed House Size

Abstract

Cembrano, Correa, Schmidt-Kraepelin, Tsigonias-Dimitriadis, and Verdugo construct randomized apportionment methods that satisfy quota compliance, population monotonicity, and ex-ante proportionality after permitting the realized house size to deviate from its target, and ask whether a similar result is possible without violating the house size. Under the formal multivalued definition of population monotonicity in their paper, we give a complete negative answer to this exact fixed-house question. Already for five states and a house of five seats, no nonempty, possibly set-valued apportionment method can satisfy both quota and population monotonicity. Consequently, no randomized method can simultaneously have exact house size and quota ex post, population monotonicity ex post, and exact quotas in expectation. The proof uses one base population profile and two comparison-profile templates. Every decisive population-ratio comparison is strict, so the equal-ratio exception in the multivalued monotonicity axiom cannot apply. Thus the result is a solution, rather than merely partial progress, for the literal formalization of the question; it does not address weaker approximate or purely stochastic notions of monotonicity.

1 Introduction and status of the question

For a population vector $\mathbf{p} = (p_1, \dots, p_n)$ and a target house size H , the quota of state i is

$$q_i(\mathbf{p}, H) = \frac{Hp_i}{\sum_{r=1}^n p_r}.$$

Cembrano et al. [1] introduce randomized fixed-divisor methods whose realized total number of seats may differ from H . Their construction satisfies quota compliance and population monotonicity ex post and exact proportionality in expectation. In their discussion they ask whether an analogous result can hold without violating the house size.

The natural exact formulation asks for a randomized method whose realized allocation always sums to H , is quota-compliant, is population-monotone ex post, and has expectation equal to the quota vector. We resolve this formulation in the negative. In fact, the expectation requirement plays no role: exact house size, quota, and population monotonicity are already incompatible.

Solution status. Theorem 2 is a complete negative solution to the exact fixed-house problem under the definitions of Cembrano et al.; it is not only partial progress. The source sentence is informal, however, so the theorem should not be read as excluding every possible approximate interpretation of “a similar result.” In particular, it does not rule out relaxations of quota, approximate house-size feasibility, monotonicity only in expectation, or other weaker stochastic monotonicity conditions.

The broad incompatibility between quota and population monotonicity is classical. Gözl, Peters, and Procaccia [2] gave a five-state impossibility for single-valued apportionment solutions without neutrality or order-preservation assumptions. The point requiring care here is that Cembrano et al.

permit multivalued methods and include a special alternative when the relevant population ratio is unchanged. The witness below uses only strict ratio changes and therefore applies directly to that multivalued axiom.

2 Model

Fix $n \geq 2$ and write $[n] = \{1, \dots, n\}$. A fixed-house, possibly set-valued apportionment method is a map

$$f : \mathbb{N}^n \times \mathbb{N} \longrightarrow 2^{\mathbb{N}_0^n} \setminus \{\emptyset\}$$

such that, for every $(\mathbf{p}, H)$ and every $\mathbf{x} \in f(\mathbf{p}, H)$,

$$\sum_{i=1}^n x_i = H.$$

The quota vector is $\mathbf{q}(\mathbf{p}, H)$ with coordinates

$$q_i(\mathbf{p}, H) = \frac{H p_i}{P}, \quad P = \sum_{r=1}^n p_r.$$

The method is *quota-compliant* if, for every instance, every $\mathbf{x} \in f(\mathbf{p}, H)$, and every state i ,

$$x_i \in \{\lfloor q_i(\mathbf{p}, H) \rfloor, \lceil q_i(\mathbf{p}, H) \rceil\}.$$

We next reproduce the population-monotonicity condition relevant to the question. A method is *population-monotone* in the sense of Cembrano et al. if, for every population vectors $\mathbf{p}, \mathbf{p}'$, house sizes H, H' , outcomes $\mathbf{x} \in f(\mathbf{p}, H)$ and $\mathbf{x}' \in f(\mathbf{p}', H')$, and states i, j , the implication

$$\frac{p'_i}{p'_j} \geq \frac{p_i}{p_j}$$

forces at least one of the following alternatives:

- (i) $x_i \leq x'_i$;
- (ii) $x_j \geq x'_j$;
- (iii) $p'_i/p'_j = p_i/p_j$ and $\mathbf{x}' \in f(\mathbf{p}, H)$.

The third alternative is the equal-ratio exception that is absent from the usual single-valued strict-ratio formulation.

We will use only the following immediate consequence.

Lemma 1 (Strict-ratio consequence). *If f is population-monotone and*

$$\frac{p'_i}{p'_j} > \frac{p_i}{p_j},$$

then no $\mathbf{x} \in f(\mathbf{p}, H)$ and $\mathbf{x}' \in f(\mathbf{p}', H')$ can satisfy simultaneously

$$x_i > x'_i \quad \text{and} \quad x_j < x'_j.$$

Proof. Under a strict ratio increase, alternative (iii) in the definition is impossible. The displayed seat inequalities negate alternatives (i) and (ii), respectively. $\square$

It is useful to emphasize that Lemma 1 is weaker than the full axiom. Therefore, an impossibility proved using only this lemma automatically applies to the full population-monotonicity definition.

3 A five-state fixed-house impossibility

Theorem 2. *For $n = 5$ and fixed house size $H = 5$, there is no nonempty set-valued apportionment method that is both quota-compliant and population-monotone. Indeed, quota is incompatible even with the weaker strict-ratio condition in Lemma 1.*

Proof. Suppose for contradiction that such a method f exists. Let

$$M = \{2, 3, 4, 5\}$$

denote the four minor states, and begin with the population profile

$$\mathbf{p}^A = (53, 6, 6, 6, 6). \quad (1)$$

Its total population is 77, so at house size five the quotas are

$$\mathbf{q}^A = \left(\frac{265}{77}, \frac{30}{77}, \frac{30}{77}, \frac{30}{77}, \frac{30}{77} \right). \quad (2)$$

Choose an arbitrary $\mathbf{x} \in f(\mathbf{p}^A, 5)$. Quota compliance gives

$$x_1 \in \{3, 4\}, \quad x_r \in \{0, 1\} \quad (r \in M).$$

We treat the two possible values of x_1 .

Case 1: $x_1 = 4$. Because the house contains five seats, exactly one minor state receives one seat. Let $k \in M$ be the unique state with $x_k = 1$; thus $x_r = 0$ for every $r \in M \setminus \{k\}$.

Define a comparison profile $\mathbf{p}^{B,k}$ by

$$p_1^{B,k} = 9, \quad p_k^{B,k} = 3, \quad p_r^{B,k} = 1 \quad (r \in M \setminus \{k\}). \quad (3)$$

The total population is 15, and hence the quotas at house size five are

$$q_1^{B,k} = 3, \quad q_k^{B,k} = 1, \quad q_r^{B,k} = \frac{1}{3} \quad (r \in M \setminus \{k\}).$$

Let $\mathbf{y} \in f(\mathbf{p}^{B,k}, 5)$ be arbitrary. Quota forces

$$y_1 = 3 \quad \text{and} \quad y_k = 1.$$

The remaining three minor states must receive exactly one seat in total. Therefore there exists $r \in M \setminus \{k\}$ with $y_r = 1$. For this state,

$$x_1 = 4 > 3 = y_1, \quad x_r = 0 < 1 = y_r.$$

On the other hand, the relative population changes strictly in favor of state 1 over state r :

$$\frac{p_1^{B,k}}{p_r^{B,k}} = 9 > \frac{53}{6} = \frac{p_1^A}{p_r^A},$$

where strictness follows from $54 > 53$. This contradicts Lemma 1 with the ordered pair of states $(1, r)$.

Case 2: $x_1 = 3$. Now the minor states receive two seats in total. Since each receives either zero or one, the set

$$S = \{r \in M : x_r = 1\}$$

has cardinality $|S| = 2$.

Define a comparison profile $\mathbf{p}^{C,S}$ by

$$p_1^{C,S} = 88, \quad p_r^{C,S} = 10 \quad (r \in S), \quad p_r^{C,S} = 1 \quad (r \in M \setminus S). \quad (4)$$

Its total population is 110, and its quotas at house size five are

$$q_1^{C,S} = 4, \quad q_r^{C,S} = \frac{5}{11} \quad (r \in S), \quad q_r^{C,S} = \frac{1}{22} \quad (r \in M \setminus S).$$

Let $\mathbf{y} \in f(\mathbf{p}^{C,S}, 5)$ be arbitrary. Quota forces $y_1 = 4$, so all four minor states receive only one seat in total. Since S contains two states, there exists $s \in S$ with $y_s = 0$. By the definition of S , $x_s = 1$. Thus

$$x_s = 1 > 0 = y_s, \quad x_1 = 3 < 4 = y_1.$$

Meanwhile the relative population changes strictly in favor of state s over state 1:

$$\frac{p_s^{C,S}}{p_1^{C,S}} = \frac{10}{88} = \frac{5}{44} > \frac{6}{53} = \frac{p_s^A}{p_1^A}.$$

Indeed, $5 \cdot 53 = 265 > 264 = 6 \cdot 44$. This contradicts Lemma 1 with the ordered pair $(s, 1)$.

Both possible values of x_1 lead to contradictions. Hence no such method f exists. $\square$

Remark 3 (A fixed finite witness). No neutrality assumption is hidden in the proof. The comparison profile is indexed by the actual outcome at $\mathbf{p}^A$: there are four labeled profiles $\mathbf{p}^{B,k}$, one for each $k \in M$, and six labeled profiles $\mathbf{p}^{C,S}$, one for each two-element subset $S \subseteq M$. Thus the contradiction is witnessed on a fixed collection of eleven profiles: the base profile $\mathbf{p}^A$, four profiles of type B , and six profiles of type C .

Remark 4 (Why strictness matters). The inequalities

$$9 > \frac{53}{6} \quad \text{and} \quad \frac{5}{44} > \frac{6}{53}$$

are essential for applying the argument directly to the multivalued axiom of Cembrano et al. Because the ratios are unequal, the axiom's equal-ratio alternative can never rescue either comparison.

4 Randomized methods

The randomized model in Cembrano et al. draws an entire apportionment method independently of the instance and then uses a tie-breaking distribution when the realized method has multiple outcomes. Quota and population monotonicity are ex-post requirements: methods violating either property must be assigned probability zero.

Corollary 5. *There is no randomized apportionment method that, for every instance, simultaneously satisfies*

- (a) *exact house size $\sum_i X_i = H$ almost surely;*

- (b) *quota almost surely;*
- (c) *population monotonicity ex post in the sense of Cembrano et al.; and*
- (d) *ex-ante proportionality $\mathbb{E}[X_i] = q_i$ for every state i .*

The impossibility remains true after deleting condition (d).

Proof. If a randomized method satisfied (a)–(c), then with probability one its realized deterministic, possibly set-valued method would have exact house size, satisfy quota, and be population-monotone. Restricting such a realization to five states and house size five would contradict Theorem 2. Hence no randomized method satisfies (a)–(c), and adding the expectation condition (d) cannot restore feasibility. $\square$

Proposition 6 (Pointwise almost-sure formulation). *The same conclusion holds if quota and each pairwise population-monotonicity constraint are initially required only on separate probability-one events, provided all allocations are generated from one common random seed.*

Proof. Population vectors, house sizes, states, outcome coordinates, and ordered pairs of profiles form countable sets. There are therefore only countably many quota and pairwise monotonicity constraints. Intersecting their probability-one events yields a probability-one event on which every constraint holds simultaneously. Each seed in this event induces a fixed-house deterministic realization satisfying quota and population monotonicity on all instances, contradicting Theorem 2. $\square$

5 Scope and relation to prior impossibilities

Theorem 2 settles the exact formal question because the desired randomized construction would require ex-post realizations that the theorem shows do not exist. The obstruction is stronger than a failure of ex-ante proportionality: it occurs before expectations enter the model.

The broad message that quota conflicts with population monotonicity is not new. In particular, Gözl, Peters, and Procaccia [2] prove an incompatibility for single-valued solutions without imposing neutrality. The present witness is tailored to the formalism of Cembrano et al. in two ways: (i) it permits the method to be set-valued, and (ii) both decisive ratio comparisons are strict, so the exceptional equal-ratio clause cannot apply. Thus the result should be viewed as a direct completion of the exact fixed-house question in that formalism, rather than as the discovery of the general quota-versus-monotonicity phenomenon.

The proof does not establish that five states or five seats are minimal. Nor does it preclude methods satisfying weakened notions such as expected population monotonicity, marginal stochastic monotonicity, approximate quota, or a small ex-post deviation from the house size.

References

- [1] Javier Cembrano, José Correa, Ulrike Schmidt-Kraepelin, Alexandros Tsigonias-Dimitriadis, and Víctor Verdugo. New combinatorial insights for monotone apportionment. *Mathematics of Operations Research*, 2025. doi:10.1287/moor.2024.0817. Preprint: arXiv:2410.23869.
- [2] Paul Gözl, Dominik Peters, and Ariel D. Procaccia. In this apportionment lottery, the house always wins. *Operations Research*, 74(1):390–407, 2026. doi:10.1287/opre.2022.0419. Preprint: arXiv:2202.11061.